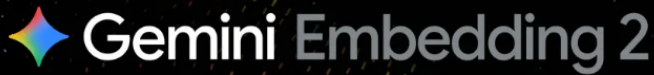


Multimodal RAG with Gemini Embedding 2.

Meet Gemini Embedding 2

Google's first natively multimodal embedding model. Released March 10, 2026.



Google just dropped **Gemini Embedding 2 -- the first embedding model that maps text, images, video, audio, and documents into a **single unified space**. No more separate pipelines. No more converting everything to text. One model. Five modalities. One vector space.**

RAG Pipelines: Before vs After

BEFORE

- ✗ Separate embeddings for text & images
- ✗ Convert PDFs to text first
- ✗ Audio needs transcription step
- ✗ Multiple indexes to maintain
- ✗ Cross-modal search = hacky workarounds



AFTER

- ✓ One model handles all 5 modalities
- ✓ PDFs embedded natively (visual + text)
- ✓ Audio embedded directly, no transcription
- ✓ Single unified vector index
- ✓ Cross-modal search just works

5 Modalities, One Space

- 01** Text -- up to 8,192 tokens, 100+ languages
- 02** Images -- up to 6 per request (PNG, JPEG)
- 03** Video -- up to 120 seconds (MP4, MOV)
- 04** Audio -- native ingestion, no transcription needed
- 05** Documents -- PDFs up to 6 pages, visual + text

Build It in 5 Lines

Embed text, images, and audio
in a single API call.

No preprocessing.

No separate models.

Just pass it in.

```
# Embed text + image + audio together
from google import genai
from google.genai import types
client = genai.Client()
result = client.models.embed_content(
    model="gemini-embedding-2-preview",
    contents=[
        "What is multimodal RAG?",
        types.Part.from_bytes(image,
            "image/png"),
        types.Part.from_bytes(audio,
            "audio/mpeg"),
    ]
)
```

Multimodal RAG Flow

Ingest: Text + Images + Audio + Video + PDFs



Embed: Gemini Embedding 2 (unified space)



Store: Vector DB (Chroma / Weaviate / Qdrant)



Query: User asks question (any modality)



Retrieve: Top-k nearest neighbors



Generate: Gemini Pro answers with context

Why Matryoshka Matters

- Default output: 3,072 dimensions
- Scale down to 1,536 or 768 dynamically
- Smaller = faster search, lower storage cost
- Larger = higher accuracy
- Same model, flexible tradeoff
- No retraining needed

8 Built-in Task Types

- 01** RETRIEVAL_QUERY -- search queries for RAG
- 02** RETRIEVAL_DOCUMENT -- index docs for retrieval
- 03** SEMANTIC_SIMILARITY -- recommendations, dedup
- 04** CLASSIFICATION -- sentiment, spam detection
- 05** CLUSTERING -- organize docs, anomaly detection
- 06** QUESTION_ANSWERING -- chatbots, QA systems
- 07** CODE_RETRIEVAL_QUERY -- find code blocks
- 08** FACT_VERIFICATION -- automated fact-checking

Plug Into Your Stack

- LangChain + LlamaIndex
- Haystack
- Weaviate, Qdrant, ChromaDB
- Google Vector Search
- BigQuery, Pinecone
- Vertex AI RAG Engine

Model ID: gemini-embedding-2-preview

Available via Gemini API + Vertex AI



**The best RAG pipelines don't
convert everything to text first.
They understand each modality
natively.**

Key Takeaway

- One embedding model for ALL modalities
- Interleaved input (image + text in one call)
- Matryoshka dimensions (3072 / 1536 / 768)
- 8 task types for optimized retrieval
- Drop-in replacement for text-only embeddings
- RAG just got multimodal. For real.

Follow for More!



Shaïd

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#GeminiAPI #MultimodalRAG #GenAI #Embeddings #AI